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Environmental Factors Associated With Risk of Crohn's Disease Development in the Crohn's and Colitis Canada - Genetic, Environmental, Microbial Project

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BACKGROUND & AIMS: **To date, it is unclear how environmental factors influence Crohn's disease (CD) risk and how they interact with biological processes. This study investigates the association between environmental exposures and CD risk and evaluates their association with pre-disease biomarkers.**

METHODS: **We studied 4289 healthy first-degree relatives (FDRs) of patients with CD from the Crohn's and Colitis Canada - Genetic, Environmental, Microbial (CCC-GEM) project. Regression models identified environmental factors associated with future CD onset and their association with pre-**

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Abbreviations used in this paper: CD, Crohn's disease; CI, confidence interval; CCC-GEM project, Crohn's and Colitis Canada - Genetic, Environmental, Microbial project; FCP, fecal calprotectin; FDRs, first-degree relatives; HR, hazard ratio; IBD, inflammatory bowel disease; IQR, interquartile range; LMR, lactulose to mannitol ratio; OR, odds ratio; SD, standard deviation.

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disease biological factors, including altered intestinal permeability measured by urinary fractional excretion of lactulose to mannitol ratio (LMR); gut inflammation via fecal calprotectin (FCP) levels; and fecal microbiome composition through 16S rRNA sequencing.

RESULTS: **Over a 5.62-year median follow-up, 86 FDRs developed CD. Living with a dog between ages 5 and 15 (hazard ratio [HR], 0.62; 95% confidence interval [CI], 0.40–0.96; $P = .034$), and living with a large family size in the first year of life (HR, 0.43; 95% CI, 0.21–0.85; $P = .016$) were associated with decreased CD risk, whereas having a bird at the time of recruitment (HR, 2.78; 95% CI, 1.36–5.68; $P = .005$) was associated with an increased CD risk. Furthermore, living with a dog was associated with reduced LMR, altered relative abundance of multiple bacterial genera, and increased Chao1 diversity, whereas bird owners had higher FCP levels. Large family during participants' first year of life was associated with altered microbiota composition without affecting FCP or LMR.**

CONCLUSION: **This study identifies environmental variables associated with CD risk. These variables were also associated with altered barrier function, subclinical inflammation, and gut microbiome composition shifts, suggesting potential roles in CD pathogenesis.**

Keywords: Crohn's Disease; Environmental Factors; Gut Inflammation; Gut Microbiome; Intestinal Permeability.

Crohn's disease (CD) is a subtype of inflammatory bowel disease (IBD) with a growing incidence worldwide. The leading hypothesis for the pathogenesis of CD involves the interplay between environmental factors and the gut microbiota in genetically susceptible individuals, resulting in an abnormal mucosal immune response and compromised epithelial barrier function.[1] The identification of more than 200 genetic loci associated with IBD, and the known increased incidence of CD in subjects with a family history support a genetic predisposition for CD.[2-5] However, the lack of complete concordance of CD among monozygotic twins,[6] and the clustering of CD incidence in siblings of multiplex families with 2 or more IBD cases suggests that environmental exposure is an important contributor to the disease.[7] Moreover, a population-based retrospective cohort study of all residents of Ontario, Canada during the period of 1994 to 2009 indicated that individuals who arrived at a younger age in Canada (which has a high incidence of IBD), are at increased IBD risk even if they immigrated from a region with a low incidence of IBD.[8] In addition, the rapid increase of incidence of IBD in newly industrialized countries in Asia and South America,[9-11] all highlight the important role of environmental factors in the development of CD.

Environmental factors, such as smoking, large families, urban environments, and exposure to pets, have been shown to be associated with the risk of CD development.[12-20] However, most of these studies were based on a retrospective study design, which makes it challenging to understand when and how environmental factors trigger the biological changes that lead to disease. Indeed, environmental exposures may influence the development of immune-mediated diseases by impacting the function of the gut microbiome[21,22] or intestinal physiology (eg, as reflected by alterations in permeability).[23,24] Previously, pre-disease intestinal microbiota composition,[25] intestinal barrier function, and gut inflammation[26] were shown to be associated with future onset of CD. However, little is known about how specific environmental factors may contribute to these pathophysiologic mechanisms.

In this study, we analyzed a multi-centered prospective cohort to investigate the association between environmental factors and the future risk of CD onset and assessed their association with other pre-disease biomarkers.

Materials and Methods

Subject Recruitment

In this study cohort, 4540 asymptomatic first-degree relatives (FDRs) of patients with CD completed an Environmental Risk Assessment questionnaire as part of the Crohn's and Colitis Canada - Genetic, Environmental, Microbial (CCC-GEM) project cohort. Participants who were healthy FDRs (siblings or offspring) of patients with CD between the age of 6 and 35 years were recruited from 2008 to 2021 and were enrolled if they had no history or symptoms of IBD and no other gastrointestinal disease. Demographic information was recorded (Supplementary Appendices 1 and 2) as previously described.[27] All participants and/or their guardians gave written informed consent to participate in the study. The study was approved by the Mount Sinai Hospital Research Ethics Board (Toronto – Managing center) and the local recruitment centers (see the list in the CCC-GEM Project Research Consortium section). When a participant disclosed that they were diagnosed with CD, this was confirmed by their treating physician and the site director (Supplementary Appendix 3). If a diagnosis could not be confirmed by the physicians or site director, the given participant was excluded from the analysis (Supplementary Figure 1).

Assessment of Environmental Factors

The Environmental Risk Assessment is a questionnaire based on previous publications,[13] and is designed to assess potential environmental triggers of CD. Specifically, the questionnaire queries 66 possible environmental exposures as well as demographic characteristics (Supplementary Appendix 4). The section headings include: Part A – Background Information; Part B – Cultural/Ethnic Background; Part C – Smoking History; Part D – Medical History; Part E – Family History; Part F – Environmental History, and Part G – Pet History. The "Environmental History" section includes multiple variables from different age groups of exposure (<1, 2–4, 5–15, 16–24, 25–30 years old), whereas the "Pet History" section, in addition to including variables from the aforementioned age groups, also captures the current exposure situation (at the time of recruitment). Specifically, the answers to the questions include 51 categorical, 3 numerical, and 12 descriptive answers. Descriptive questions were excluded from the analysis. We adopted a unified standard for quality control, which involved excluding questions with answers categorized as 'not applicable,' 'don't know,' or having 'missing values' exceeding 5%. Considering that 43% of participants in the GEM project were under 16 years old (1834 of 4289 participants), responses from those under 16 to questions for the 16–24 and 25–30 age groups were marked as 'not applicable.' As a result, all questions for these age groups were excluded. In the end, 65 variables were included in the analysis (Supplementary Table 1; Supplementary Figure 2). For the multi-category variables, given the limited number of participants in certain categories, we dichotomized them to prevent convergence issues in regression models. For variables with missing values less than 5%, multiple imputation[28,29] was used to impute missing and "don't know" values (MICE package v 3.14.0). Continuous missing data were imputed with the average of the prediction results of imputation models using default parameters. Categorical missing data were interpolated with the mode of the prediction results of imputation models using default parameters. R software (version 4.0.4) was used for statistical analysis.

Sample Collection

The participants' stool samples were collected as previously described.[26] We analyzed fecal calprotectin (FCP) from stool samples collected at recruitment, as previously described.[26] DNA from stool was sequenced for 16S ribosomal RNA, as previously described.[30] The intestinal barrier function was measured, as previously described.[26] For details, see Supplementary Methods.

Statistical Analysis of the Association Between Environmental Risk Assessment and CD Onset

To address the potentially highly correlated variables within the Environmental Risk Assessment, we employed the hierarchical clustering and Spearman correlation method for feature selection, utilizing the default parameters from the *scipy.cluster.hierarchy* Python package v1.6.3.[31] We then followed a 3-step processing to identify environmental factors associated with CD onset (See Supplementary Methods).

Statistical Analysis of the Association Between pre-CD Environmental Factors and Biomarker of Intestinal Permeability and Biomarker of Subclinical Intestinal Inflammation

The role of pre-CD environmental factors in CD pathogenesis was further investigated for their association with pre-CD biomarkers, specifically focusing on gut permeability and gut inflammation (See Supplementary Material).

What You Need to Know

Background

Epidemiological studies suggest a critical role of environmental factors on risk for developing Crohn's disease (CD). How these exposures are related to risk is unknown and their impact on other biomarkers of CD risk is unknown.

Findings

The exploration of a number of environmental exposures has shown that living with a dog as a pet and living in a large family early in life decreases CD risk and are associated with altered microbiota composition and gut permeability.

Implications for patient care

Biomarkers of CD risk associated with environmental factors suggest possible markers of environmental exposures. Adopting a lifestyle comprised of protective environmental factors could be used as a preventive measure to reduce risk of CD.

Results

Baseline Characteristics of Participants

The cohort analyzed included 4203 participants who remained asymptomatic as of February 2021, and 86 participants who developed CD (referred to as pre-CD) during the median 5.6-year follow-up period (interquartile range [IQR], 3.42–8.67) (Supplementary Figure 1; Table 1). The cohort's median recruitment age was 17.0 years (IQR, 6–35 years), with 47% being male. Participants who remained healthy were older at recruitment than those who developed CD ($P = .04$), and there were no significant differences in terms of sex, self-reported ethnicity, or country of birth.

Table 1. Demographic Data of the Cohort Included in the Study

	Total cohort	Remain healthy	Developed CD	*P*-value
Participants	N = 4289	n = 4203	n = 86	–
Sex[a]				.543
Female	2280 (53.2)	2231 (53.1)	49 (57.0)	
Male	2009 (46.8)	1972 (46.9)	37 (43.0)	
Age at recruitment, *years*[c]	17.00 (12.00–24.00)	17.00 (12.00–24.00)	14.50 (12.00–20.75)	.040[e]
Race and ethnicity[b]				1.0
White	3903 (91.0)	3824 (91.0)	79 (91.9)	
Other[d]	386 (9.0)	379 (9.0)	7 (8.1)	
Geography of birth[a]				.056
North America	2907 (67.8)	2840 (67.6)	67 (77.9)	
Non-North America	1382 (32.2)	1363 (32.4)	19 (22.1)	
Time of follow-up/time to diagnosis, *years*	5.62 (3.42–8.67)	5.68 (3.47–8.73)	2.79 (1.34–4.64)	–

Note: Data are presented as number (%) or median (interquartile range).
Note: All *P* values are obtained by comparing the participant who remained healthy against the participant that developed CD.
CD, Crohn's disease.
[a]*P* values are obtained using the χ^2 test.
[b]*P* values are obtained using the Fisher exact test.
[c]*P* values are obtained using the Mann-Whitney *U* test.
[e]Indicates a significant difference between remain healthy and developed CD group ($P < .05$).

Association of Environmental Exposures With CD onset

Sixty-five variables were assessed for association with CD risk (Supplementary Table 2). A total of 19 variables were selected from 65 variables from the Environmental Risk Assessment after feature selection (see Methods). Four of the 19 variables were significantly associated with CD (Supplementary Table 3). Specifically, participants who were living with a dog from ages 5 to 15 years were less likely to develop CD compared with subjects who were not living with a dog in this age group (hazard ratio [HR], 0.61; 95% confidence interval [CI], 0.39–0.94; $P = .025$). Participants who developed CD were less likely to have a larger family during their first year of life (defined as 3 or more individuals in a given household) than individuals who remained healthy (HR, 0.43; 95% CI, 0.22–0.87; $P = .019$). Participants who were living with birds at the time of recruitment had an increased risk of developing CD compared with participants who were not living with birds (HR, 2.84; 95% CI: 1.37–5.9; $P = .005$) (Figure 1). Additionally, the demographic feature indictive of participants who have a sibling with CD (as compared with offspring) was associated with an increased risk of CD during the follow-up period (HR, 2.02; 95% CI, 1.15–3.56; $P = .014$). In further analyses, we adjusted for identified demographic features, comparing those with a sibling with CD vs offspring with CD, to assess the robustness of the association between environmental features and CD (Supplementary Table 4, 5, and 6; Supplementary Material). To better understand the effect of the timing of exposure to a given environmental factor, we further assessed the association between pre-CD related environmental factors during other exposure age groups (<1, 2–4, 5–15 and at recruitment) with future CD risk (Supplementary Tables 5–8; Supplementary Material).

Associations of CD Risk Environmental Exposures and Gut Permeability

A total of 3341 participants had Environmental Risk Assessment and gut permeability measured at recruitment. Of these, 851 (25.5%) exhibited abnormal gut permeability (lactulose to mannitol ratio [LMR] >0.025). Further analysis revealed that participants who lived with a dog from ages 5 to 15 had a significantly lower LMR compared with those who did not live with a dog (coefficient ± standard deviation [SD]: -0.02 ± 0.01; $P = .002$) (Supplementary Table 11). This finding was consistent when LMR was assessed as a dichotomous variable (See Methods), with an odds ratio (OR) of 0.77 (95% CI, 0.65–0.90; $P = .001$) (Figure 2). We then assessed whether living with a dog in different age groups was consistently associated with reduced LMR (<1, 2–4, and at recruitment). Intriguingly, participants who had lived with a dog at any age had a lower incidence of abnormal gut permeability and lower LMR values compared with those who had not (Supplemental Figures 3 and 4; Supplemental Table 12). We did not observe any statistically significant associations between continuous or dichotomous LMR with the other environmental factors that were significantly associated with CD (Figure 2; Supplementary Table 11).

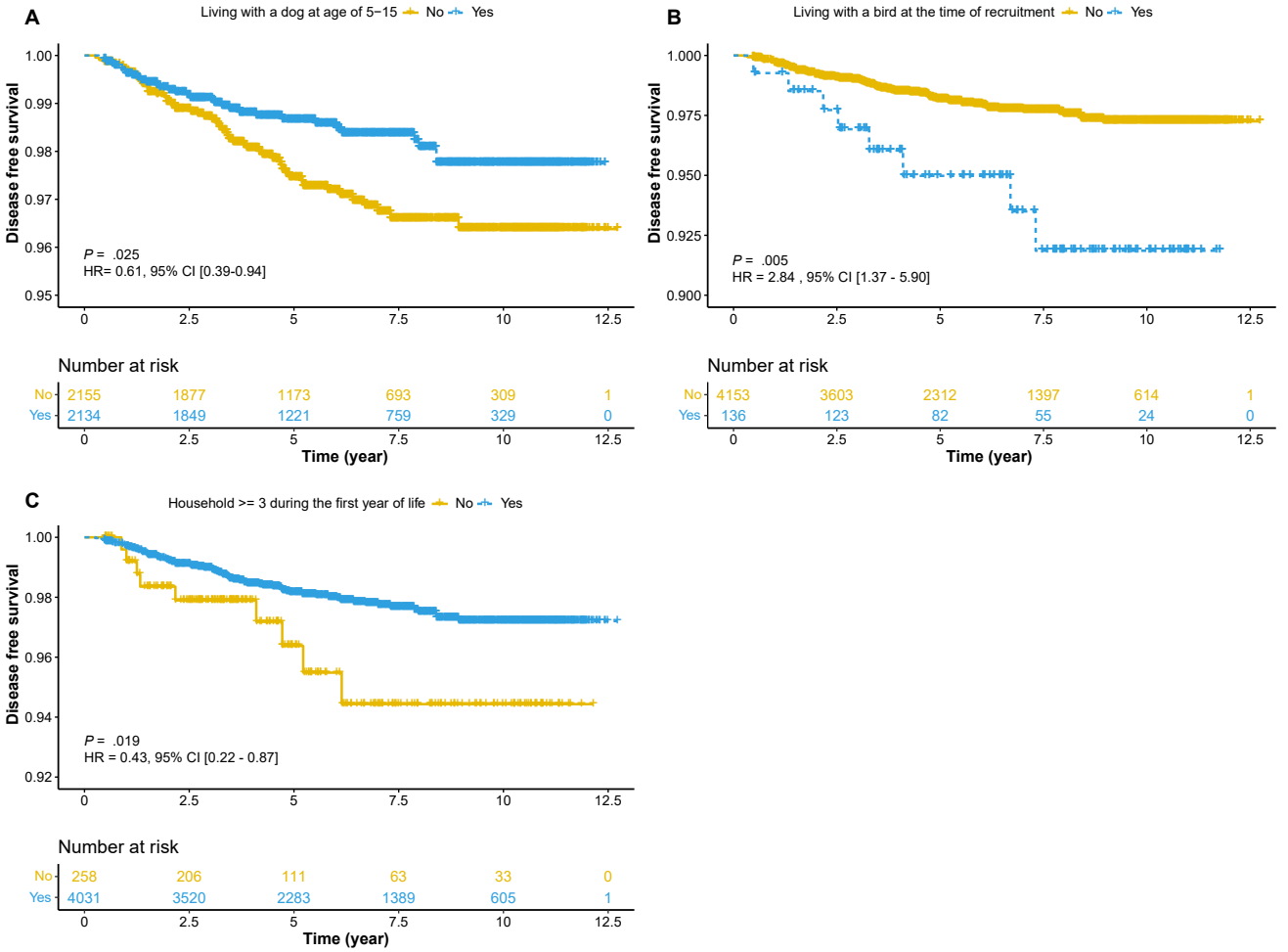


Figure 1. Environmental exposures independently associated with the risk of onset of Crohn's disease. Kaplan-Meier plot of the comparison of participants exposed or not exposed to living with a dog at the age of 5-15 (*A*); living with a bird at the time of recruitment (*B*); and living in a household ≥3 during the first year of life (*C*). The x-axis represents the time (in years) between the time of answering the questionnaire and the last follow-up of the participants. The y-axis represents the incidence-free probability of developing CD. The *blue line* represents the participants exposed to environmental variables, and the *yellow line* represents the participants who were not exposed to environmental variables. The *p*-value was based on multivariate Cox proportional hazards analysis (see Methods section).

Association of CD Risk Environmental Exposures and Subclinical Gut Inflammation

A total of 3727 participants had Environmental Risk Assessment and FCP data assessed. We found that 772 of those participants (20.7%) exhibited subclinical inflammation at recruitment (defined as FCP >100 μg/g). When FCP was assessed as a dichotomized variable, participants currently living with a bird were more likely to have subclinical inflammation compared with other participants (OR, 2.04; 95% CI, 1.33–3.11; P = .001) (Figure 3). Similar results were also shown using FCP as a continuous variable (coefficient (±SD): 68.88 ± 31.56; P = .029) (Supplementary Table 13). No association was found between FCP (assessed either as continuous or dichotomized variable) and living with a dog from ages 5 to 15 or having a large family household during the first year of life (Figure 3; Supplementary Table 13).

We then assessed whether living with a bird in different age groups (>1, 2–4, and 5–15 years) was consistently associated with FCP as a continuous or dichotomized variable. We found that the participants living with birds from 5–15 years of age were more likely to have elevated FCP (OR, 1.29; 95% CI, 1.02–1.63; P = .035) (Supplementary Figure 5). However, when FCP was assessed as a continuous variable, the significant difference no longer exists (coefficient (±SD): 12.87 ± 11.86; P = .28) (Supplementary Table 10). Finally, living with birds during the first year of life or in the 2 to 4 age group was not associated with either continuous or dichotomous FCP levels (Supplementary Figures 5 and 6; Supplementary Table 14).

Association of CD Risk Environmental Exposures With Microbiome

We assessed the CD risk-associated environmental factors with beta and alpha diversity in a subset of

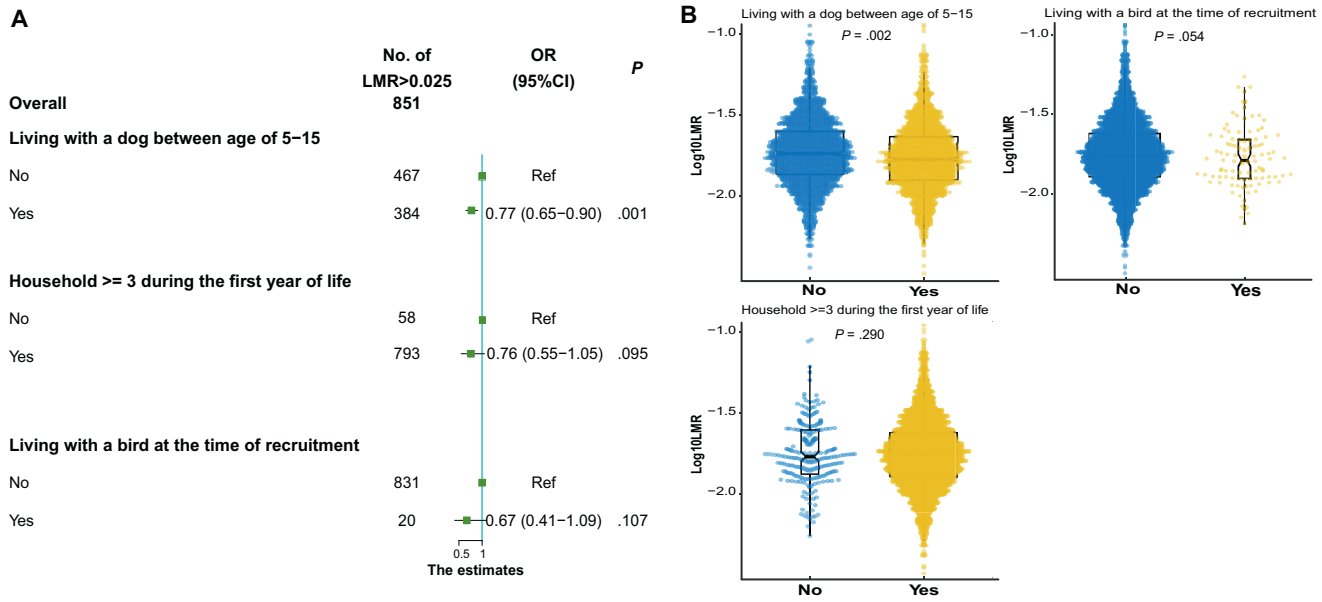


Figure 2. Association between CD-associated environmental exposures and gut permeability (LMR >0.025). (*A*) Forest plot represents an association of environmental exposures and dichotomized LMR. Squares indicate the value of OR, whereas horizontal lines represent the 95% CI of OR. Generalized estimating equation logistic regression accounts for family clustering and further adjusted for age and sex was used; (*B*) Distribution of LMR in participants exposed to and not exposed to identified environmental factors. The x-axis represents the environmental exposure of participants: No (*blue*), Yes (*yellow*). The y-axis represents the value of $\log_{10}$ LMR. The lines in the boxplot represent the first, second, and third quartiles. Each dot represents the $\log_{10}$(LMR) for a given participant. The *P*-value generated by generalized estimating equation linear regression accounts for family clustering and further adjusted for age and sex.

3693 participants who had Environmental Risk Assessment and microbiome composition measured at recruitment (Supplementary Figure 7; Supplementary Results).

We then examined associations between relative abundance of individual bacterial taxa at the genus level of taxonomy and pre-CD environmental factors (Supplementary Tables 15–18; Supplementary Figure 8; Supplementary Results). We found that participants who lived with a dog between the ages of 5 and 15 had a higher abundance of *Frisingicoccus, Incertae_Sedis, Adlercreutzia, Ruminococcaceae_UBA1819,* and *Phascolarctobacterium*

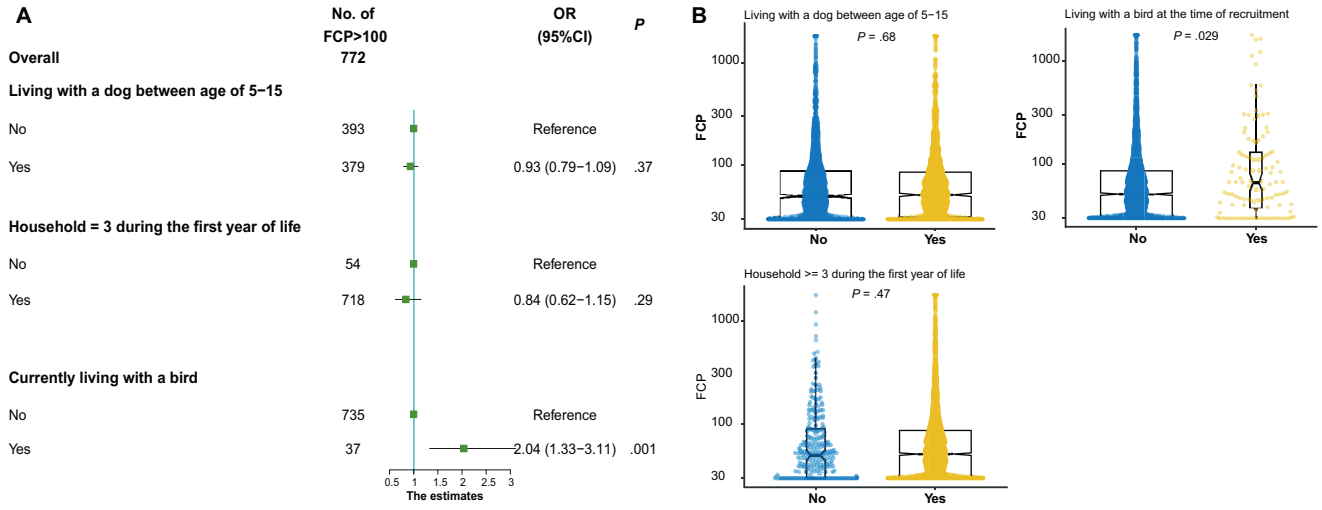


Figure 3. Association between CD-associated environmental exposures and gut inflammation (FCP >100 μg/g). (*A*) Forest plot representing the association of environmental exposures and dichotomized FCP. Squares indicate the value of OR, whereas horizontal lines represent the 95% CI of OR. Generalized estimating equation logistic regressions account for family clustering and further adjusted for age and sex were used. (*B*) Distribution of FCP in participants exposed and not exposed to CD-associated environmental factors. The x-axis represents the given environmental exposure of participants: No (*blue*), Yes (*yellow*). The y-axis represents the value of FCP; y-axis is $\log_{10}$ transformed for visualization. The lines in the boxplot represent the first, second, and third quartiles. Each dot represents the FCP for a given participant. The *P*-value generated by generalized estimating equation linear regression accounts for family clustering and further adjusted for age and sex.

($q < 0.050$), and a lower abundance of *Senegalimassilia*, *Lactobacillus, Dialister,* and *Lachnospiraceae_UCG-004* ($q < 0.05$) compared with those who did not live with a dog during this period.

Discussion

In this prospective at-risk cohort of 4289 healthy FDRs of patients with CD, we identified several environmental factors associated with CD onset. Specifically, living in a larger household in the first year of life, as well as living with a dog between the ages of 5 and 15, were both identified as protective factors against the risk of CD; concurrently, currently living with birds has been identified as a risk factor. Our analysis revealed a protective trend of living with dogs that transcends the age of exposure, suggesting that dog ownership could confer health benefits in reducing the risk of CD. Additionally, we observed that living with birds might increase the risk of CD onset; however, these findings should be interpreted with caution due to the relatively small number of participants who lived with birds (176 individuals aged 2–4; 529 individuals aged 5–15; 136 individuals at the time of recruitment). Previous research has established that childhood exposure to pets and farm animals can affect the future risk of IBD.[12,14,15,32] Our study also found that living in a large family during the first year of life is significantly associated with the future onset of CD, aligning with prior research that indicates that a larger family size in the first year of life can reduce the risk of developing IBD.[15] In summary, our research has identified multiple environmental factors associated with the risk of CD, based on the timing of exposure, in a cohort of FDRs of patients with CD (See Supplementary Discussion).

Our study has several limitations. First, we recognize that questionnaire-based assessments of environmental exposure, although cost-effective tools for collecting large-scale data, can be subject to recall bias.[33] In addition, questionnaires are intrinsically limited to environmental questions asked and cannot provide data on novel or unknown environmental factors that may be associated to CD risk. The use of untargeted and less biased methods such as metabolomics or exposomics might hold the key to discovering novel environmental risk factors. Secondly, the association of environmental factors with biological markers in our study were based on a single measurement taken at the time of recruitment. This approach did not consider the potential variability over time, although it is worth noting that previous studies have shown that the human microbiome and gut permeability are relatively stable after the early stages of life.[34-36] Finally, although our sample size can be considered large, some environmental variables were uncommon, such as the participants having birds as pets, and would greatly benefit from replication of our findings in other cohorts.

Conclusion

In summary, we found environmental exposures associated with CD onset. Living with a dog and having a large family during the first year of life were identified as protective exposures, whereas living with a bird increases the risk of CD. We identified potential mechanisms associated with environmental factors related to CD risk, including gut permeability, microbiome composition, and subclinical inflammation. Understanding the relationship between CD-related environmental factors and these pre-disease biomarkers may shed light on the underlying mechanisms by which environmental factors impact host health and ultimately lead to CD onset. Our findings contribute to the growing evidence supporting the potential health benefits of exposure to pets, particularly dogs, as a potential preventive strategy for individuals at risk of developing CD. Modifying environmental exposures may be an acceptable potential option for high-risk individuals.[16]

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at https://doi.org/10.1016/j.cgh.2024.03.049.

References

1. Torres J, Mehandru S, Colombel JF, Peyrin-Biroulet L. Crohn's disease. Lancet 2017;389:1741–1755.
2. Liu JZ, van Sommeren S, Huang H, et al. International Multiple Sclerosis Genetics Consortium; International IBD Genetics Consortium;. Association analyses identify 38 susceptibility loci for inflammatory bowel disease and highlight shared genetic risk across populations. Nat Genet 2015;47:979–986.
3. Moller FT, Andersen V, Wohlfahrt J, Jess T. Familial risk of inflammatory bowel disease: a population-based cohort study 1977-2011. Am J Gastroenterol 2015;110:564–571.
4. Kenny EE, Pe'er I, Karban A, et al. A genome-wide scan of Ashkenazi Jewish Crohn's disease suggests novel susceptibility loci. PLoS Genet 2012;8:e1002559.
5. Schiff ER, Frampton M, Semplici F, et al. A new look at familial risk of inflammatory bowel disease in the Ashkenazi Jewish population. Dig Dis Sci 2018;63:3049–3057.
6. Thompson NP, Driscoll R, Pounder RE, Wakefield AJ. Genetics versus environment in inflammatory bowel disease: results of a British twin study. BMJ 1996;312:95–96.
7. Spencer EA, Helmus D, Telesco S, et al; Road to Prevention Study Group. Inflammatory bowel disease clusters within affected sibships in Ashkenazi Jewish multiplex families. Gastroenterology 2020;159:381–382.
8. Benchimol EI, Mack DR, Guttmann A, et al. Inflammatory bowel disease in immigrants to Canada and their children: a population-based cohort study. Am J Gastroenterol 2015; 110:553–563.
9. Ng SC, Bernstein CN, Vatn MH, et al. Epidemiology and Natural History Task Force of the International Organization of Inflammatory Bowel Disease (IOIBD). Geographical variability and

environmental risk factors in inflammatory bowel disease. Gut 2013;62:630–649.
10. Loftus EV Jr. Clinical epidemiology of inflammatory bowel disease: incidence, prevalence, and environmental influences. Gastroenterology 2004;126:1504–1517.
11. Molodecky NA, Soon IS, Rabi DM, et al. Increasing incidence and prevalence of the inflammatory bowel diseases with time, based on systematic review. Gastroenterology 2012;142:46–54.e42; quiz: e30.
12. Timm S, Svanes C, Janson C, et al. Place of upbringing in early childhood as related to inflammatory bowel diseases in adulthood: a population-based cohort study in Northern Europe. Eur J Epidemiol 2014;29:429–437.
13. Bernstein CN, Rawsthorne P, Cheang M, Blanchard JF. A population-based case control study of potential risk factors for IBD. Am J Gastroenterol 2006;101:993–1002.
14. Radon K, Windstetter D, Poluda AL, et al. Contact with farm animals in early life and juvenile inflammatory bowel disease: a case-control study. Pediatrics 2007;120:354–361.
15. Amre DK, Lambrette P, Law L, et al. Investigating the hygiene hypothesis as a risk factor in pediatric onset Crohn's disease: a case-control study. Am J Gastroenterol 2006;101:1005–1011.
16. Lopes EW, Chan SSM, Song M, et al. EPIC-IBD investigators. Lifestyle factors for the prevention of inflammatory bowel disease. Gut 2022. Online ahead of print.
17. Roberts SE, Wotton CJ, Williams JG, et al. Perinatal and early life risk factors for inflammatory bowel disease. World J Gastroenterol 2011;17:743–749.
18. van der Sloot KWJ, Weersma RK, Alizadeh BZ, Dijkstra G. Identification of environmental risk factors associated with the development of inflammatory bowel disease. J Crohns Colitis 2020;14:1662–1671.
19. Mahid SS, Minor KS, Soto RE, et al. Smoking and inflammatory bowel disease: a meta-analysis. Mayo Clin Proc 2006;81:1462–1471.
20. Russel MG, Volovics A, Schoon EJ, et al. Inflammatory bowel disease: is there any relation between smoking status and disease presentation? European Collaborative IBD Study Group. Inflamm Bowel Dis 1998;4:182–186.
21. Mohajeri MH, Brummer RJM, Rastall RA, et al. The role of the microbiome for human health: from basic science to clinical applications. Eur J Nutr 2018;57(Suppl 1):1–14.
22. Stewart CJ, Ajami NJ, O'Brien JL, et al. Temporal development of the gut microbiome in early childhood from the TEDDY study. Nature 2018;562:583–588.
23. Keirns BH, Koemel NA, Sciarrillo CM, et al. Exercise and intestinal permeability: another form of exercise-induced hormesis? Am J Physiol Gastrointest Liver Physiol 2020;319:G512–G518.
24. Ueno A, Jijon H, Traves S, et al. Opposing effects of smoking in ulcerative colitis and Crohn's disease may be explained by differential effects on dendritic cells. Inflamm Bowel Dis 2014;20:800–810.
25. Raygoza Garay JA, Turpin W, Lee SH, et al; CCC GEM Project Research Consortium. Gut microbiome composition is associated with future onset of Crohn's disease in healthy first-degree relatives. Gastroenterology 2023;165:670–681.
26. Turpin W, Lee SH, Raygoza Garay JA, et al; Crohn's and Colitis Canada Genetic Environmental Microbial Project Research Consortium. Increased intestinal permeability is associated with later development of Crohn's disease. Gastroenterology 2020;159:2092–2100.e5.
27. Turpin W, Espin-Garcia O, Xu W, et al; GEM Project Research Consortium. Association of host genome with intestinal microbial composition in a large healthy cohort. Nat Genet 2016;48:1413–1417.
28. Austin PC, White IR, Lee DS, van Buuren S. Missing data in clinical research: a tutorial on multiple imputation. Can J Cardiol 2021;37:1322–1331.
29. Buuren SV, Groothuis-Oudshoorn K. MICE: multivariate imputation by chained equations in R. J Statistic Software 2011;45(3).
30. Leibovitzh H, Lee SH, Xue M, et al; CCC GEM Project Research Consortium. Altered gut microbiome composition and function are associated with gut barrier dysfunction in healthy relatives of patients with Crohn's disease. Gastroenterology 2022;163:1364–1376.e10.
31. Virtanen P, Gommers R, Oliphant TE, et al. SciPy 1.0 Contributors. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods 2020;17:261–272.
32. Levy LC, Plevy SE. Farm animals and inflammatory bowel disease: old MacDonald had poor hygiene. Gastroenterology 2008;134:886–887; discussion: 887–888.
33. Coughlin SS. Recall bias in epidemiologic studies. J Clin Epidemiol 1990;43:87–91.
34. Łoniewska B, Adamek K, Węgrzyn D, et al. Analysis of faecal zonulin and calprotectin concentrations in healthy children during the first two years of life. An observational prospective cohort study. J Clin Med 2020;9:777.
35. Kaczmarczyk M, Löber U, Adamek K, et al. The gut microbiota is associated with the small intestinal paracellular permeability and the development of the immune system in healthy children during the first two years of life. J Transl Med 2021;19:177.
36. Faith JJ, Guruge JL, Charbonneau M, et al. The long-term stability of the human gut microbiota. Science 2013;341:1237439.

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Conflicts of interest

The authors disclose no conflicts.

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Data availability

Requests for raw and analyzed data should follow the instructions given at http://www.gemproject.ca/data-access/. The raw microbiome data are publicly accessible under Accession: PRJNA685746. All submissions will be reviewed by the GEM Project Operating Committee to ensure that the requested samples/data will not interfere in any way with the intended GEM Project analysis of the nested cohort as per the original GEM Project Study Design and is not a duplication of analysis already ongoing. Those proposals meeting this evaluation will be distributed to all members of the GEM Project Steering Committee (GPSC) for review and open discussion. This review will focus on the global scientific merit of the proposal. This review will assess the basic scientific merit and the availability of requested samples and data, ensuring there is no compromise of the original intent of the GEM project. It would be of value to contact a member of the Steering Committee who could help sponsor your application. Those projects achieving majority vote of approval at the GPSC will be informed that the GEM Project will provide a letter of support stating that the requested samples or data will be made available to the applicants once the applicant receives funding from a granting agency that applies an independent peer review process to the proposal. The criteria to be used for review of all submissions will include the "scientific relevance" of the proposal and the judged availability of biological material requested. The budget to be requested from a funding agency must allow for any expenses in processing samples or in setting up the appropriate queries of the database. The intent is to allow sufficient time for applicants to consider submission for funding opportunities.

Supplementary Results

Association of Environmental Exposures with Crohn's Disease Onset Adjusting for the Demographic Feature Sibling vs Offspring

In further analyses, we adjusted for identified demographic features, comparing those with a sibling with Crohn's disease (CD) vs offspring with CD, to assess the robustness of the association between environmental features and CD. The adjusted model revealed that living with a dog between the ages of 5 and 15 (hazard ratio [HR], 0.63; 95% confidence interval [CI], 0.41–0.98; $P = .039$), living with birds at the time of recruitment (HR, 2.86; 95% CI, 1.38–5.93; $P = .005$), and living in large family in the first year of life (HR, 0.40; 95% CI, 0.22–0.80; $P = .01$) remained significantly associated with CD onset (Supplementary Table 4).

Association of Environmental Exposures With CD Onset During Other Exposure Age Group

To better understand the effect of the timing of exposure to a given environmental factor, we further assessed the association between pre-CD related environmental factors, including living with a dog, living with a bird, and having a larger family size, during other exposure age groups (<1, 2–4, 5–15, and at recruitment) with future CD risk. Although only the group of individuals exposed during the 5- to 15-years-of-age period was statistically significant, we observed that participants living with a dog in any age group consistently showed a trend towards a decreased likelihood to develop CD (Supplemental Table 7). Similarly, there was no significant difference between participants living with a bird and future CD development in the 2 to 4 and 5 to 15 age groups, but the direction of the effects was consistent compared with participants not living with a bird (Supplemental Table 9). There were no significant differences between participants having larger households and CD development in the other age groups (Supplemental Table 10).

Association Of CD Risk Environmental Exposures With Microbiome Diversity

We found that participants who lived with a dog between the ages of 5 and 15 showed a significant increase in alpha diversity, as determined by the Chao1 index ($P = .035$). Additionally, living with a dog between the age of 5 and 15 resulted in a small but significant compositional shift (permutation analysis of variance: R^2, $9e^{-04}$; $P = .001$; 1000 permutations) (Supplementary Figure 7, *A–B*). When comparing participants with and without a large family in the first year of life, we observed a significant compositional difference (permutation analysis of variance: R^2, $5e^{-04}$; $P = .019$; 1000 permutations), but no significant difference in alpha diversity (Chao1 index: $P = .22$) (Supplementary Figure 7*C–D*). Finally, we did not find any significant differences in alpha or beta diversity when comparing participants who currently lived with a bird with those who did not at the time of recruitment (Supplementary Figure 7*E–F*).

Association Of CD Risk Environmental Exposures With Microbiome Taxonomic Composition

We found that participants who lived with a dog between the ages of 5 and 15 had a higher abundance of *Frisingicoccus, Incertae_Sedis, Adlercreutzia, Ruminococcaceae_UBA1819*, and *Phascolarctobacterium* ($q < 0.050$), and a lower abundance of *Senegalimassilia*, *Lactobacillus, Dialister,* and *Lachnospiraceae_UCG-004* ($q < 0.05$) compared with those who did not live with a dog during this period (Supplementary Table 15). No significant associations were found between any individual taxa and other pre-CD environmental factors, including living with a bird and family size (Supplementary Tables 16, 17). In analyzing the association between living with a dog at other age groups and the gut microbiome, we found significant differences in beta diversity between dog owners and non-owners across various age groups. Additionally, 6 genera (*Senegalimassilia, Lactobacillus, Incertae_Sedis, Frisingicoccus, Dialister,* and *Adlercreutzia*) were consistently different in participants who lived with a dog in any age group.

Association of Living With a Dog at Different Age Groups With Microbial Composition and Diversity

To determine if owning a dog in other age groups also affects microbiome alpha and beta diversity, we performed further analysis and found that individuals who were living with a dog between the ages of 2 and 4 years also had higher alpha diversity (Chao1 index at genus level, $P = .027$). However, we did not observe any difference in alpha diversity compared with non-dog owners in the other groups. There was a significant difference in beta diversity between participants who owned dogs and those who did not across all age groups (Supplementary Figure 7).

In our investigation of the association between living with a dog and the abundance of specific taxa across different age groups (>1, 2–4 years, and the time of recruitment), we compared the findings with those in the 5 to 15 years age group. We discovered that 6 of the 9 genera that were significant in the 5 to 15 years age group (Supplementary Table 15) were consistently different in participants who lived with a dog in any age group (*Senegalimassilia, Lactobacillus, Incertae_Sedis,*

Frisingicoccus, Dialister, and *Adlercreutzia*). However, the genera *Ruminococcaceae*_UBA1819 and *Phascolarctobacterium* were only significantly different in participants who lived with a dog after the age of 5. In contrast, *Lachnospiraceae*_UCG-004 was only significant in the 2 to 4 age group (Supplementary Table 18).

Supplementary Discussion

Exploring the Potential Mechanisms of the Association With CD Risk and Impact With CD Biomarkers

In an attempt to explore possible mechanisms that could explain the observed differences in pre-CD associated environmental factors, we separately assessed their association with intestinal permeability, gut inflammation, and gut microbiome. As mentioned above, participants living with dogs showed evidence of a lower risk of developing CD later in life. Interestingly, we found that living with a dog is associated with improved gut permeability, which is a known biomarker of future CD onset.[1] The underlying mechanism is yet to be fully elucidated. However, one potential explanation could be the increased propensity of dog owners to engage in physical activity, which has been shown to be associated with improved indicators of gut permeability.[2] Our findings, showing an association between a history of cohabitation with a dog and gut permeability, further bolster this interpretation. Additionally, exposure to green spaces has been associated with decreased inflammatory bowel disease (IBD) risk, suggesting its role as a protective factor.[3,4] Dog owners may have more frequent exposure to such green spaces, and this might explain the protective effect of owning a dog and risk of IBD development.

We found that living with a bird at the time of recruitment was associated with higher levels of fecal calprotectin (FCP). Elevated FCP in healthy first-degree relatives (FDRs) is associated with an increased risk of developing CD.[1] One might speculate that bird droppings or feather dander, which may contain microorganisms such as bacteria, fungi, and viruses, could trigger inflammation in the gut.[5] However, our study did not identify a difference in gut microbiota composition between those living with birds as a pet. It is important to note that the exact mechanisms underlying the association between living with a bird and gut inflammation/CD are not yet understood. Moreover, the sample size of our study participants who were living with a bird may limit the generalizability of our findings. Therefore, further research is needed to confirm our results and better understand the underlying mechanisms.

In our study, a large family size in the first year of life was a protective factor against CD development. One potential explanation may be increased exposure to infectious agents early in life, which could help prime the immune system to better handle bacteria that contribute to IBD.[6] Furthermore, children in larger families naturally may come into contact with more individuals and diverse environments that may broaden their encounter with various microbes. This diverse microbial exposure can strengthen the immune responses to various antigens.[7] Moreover, individuals from larger families may be exposed to a greater diversity of bacteria, promoting a "healthy" gut microbiome.[8] Our data, illustrating significant differences in gut microbiota composition between participants who lived in large families during their first year of life compared with those who did not (Supplementary Figure 8), supports the hypothesis that exposure to a large family can influence the structure of the gut microbiome.

Association of Dog Ownership and Microbiome

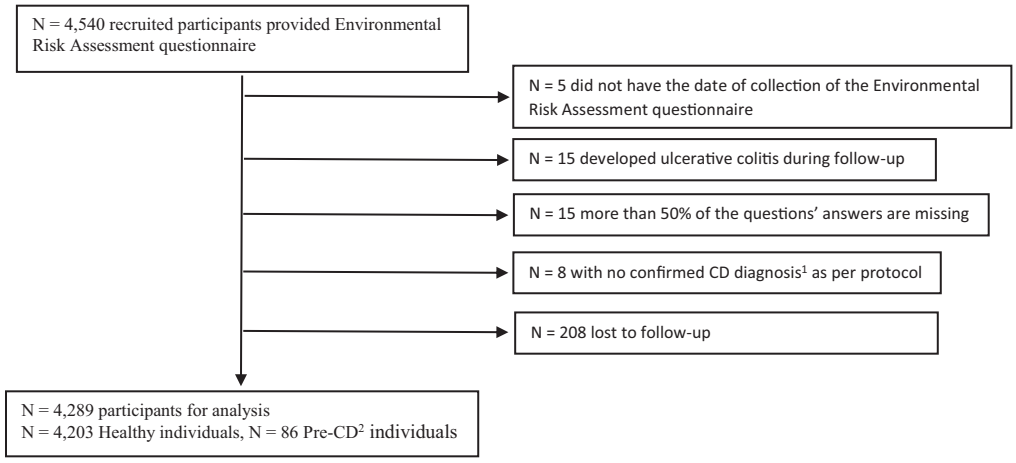
Our research also revealed a significant association between microbial composition and past dog ownership. Previous studies showed that dog ownership was associated with differences in the composition of the human skin microbiome, with dog owners having a higher abundance of certain bacterial taxa associated with skin health.[9] Another study involving 54 individuals aged over 65 found that dog owners had a higher abundance of several bacterial taxa in the intestinal tract, including *Bifidobacteriaceae* and *Ruminococcaceae*, and lower abundance of *Moracellaceae*.[10] Those taxa differ from the ones observed in our study, which might be explained by multiple factors such as sample size, demographics (age, sex), etc. Indeed, we found several other taxa associated with dog ownership, one of which is *Adlercreutzia.* We previously showed that relative abundance of *Adlecreutzia* is significantly reduced in individuals who later developed ulcerative colitis.[11] Additionally, our previous research has found that *Adlercreutzia* is significantly reduced in individuals with impaired barrier function (lactulose to mannitol ratio [LMR] >0.025), indicating its potential protective role in the onset of IBD. This also provides another potential explanation on how dog ownership might promote a healthy intestinal barrier.[12] Moreover, our data may support the so-called Hygiene Hypothesis,[7] defined as a lack of exposure to microbial diversity during early life maturation of the immune system, which can later promote uncontrolled immune responses to commensal bacteria. Indeed, a previous study found that pets, more specifically dogs, were a major contributor to microbial diversity of the indoor microbiome.[13] Another study has shown that people living with a dog had more diverse microbiomes than those without dogs.[14] These findings may help to further elucidate the mechanisms underlying the beneficial effects of pet ownership on human health.

Association of the Most Common Risk Factors Identified in Other Studies

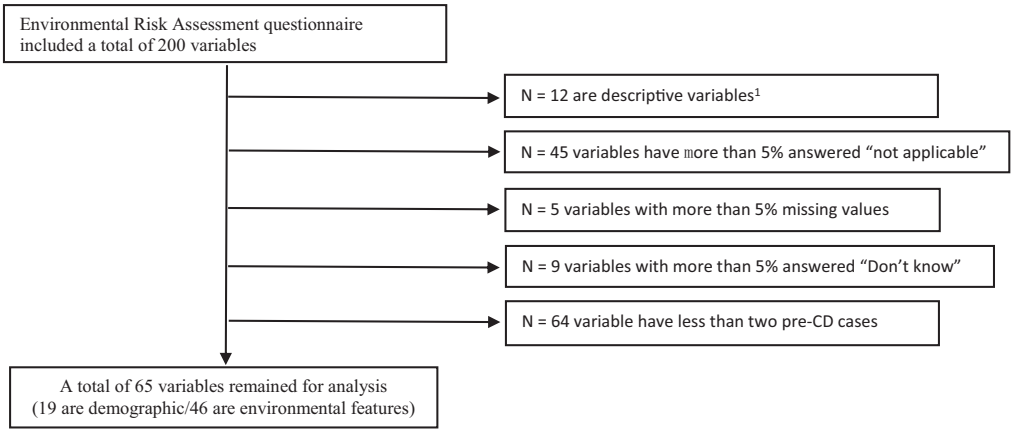
It is noteworthy that smoking, one of the most common risk factors identified in other studies,[15,16] was not identified in our cohort. This may be due to the fact the subjects in our cohort were between the ages of 6 and 35, with 52% of them younger than 18 years old. Given this age distribution, the proportion of smokers in the cohort was small (7%), which would limit our capacity to identify a significant association. However, we did find that participants with a mother who smoked during pregnancy and who lived with passive smokers had a 1.13- and 1.63-fold higher risk of developing CD than those who did not, although the *P*-value did not meet the criteria for statistical significance (Supplementary Table 2).

Supplementary References

1. Turpin W, Lee SH, Raygoza Garay JA, et al. Crohn's and Colitis Canada Genetic Environmental Microbial Project Research Consortium. Increased intestinal permeability is associated with later development of Crohn's disease. Gastroenterology 2020;159:2092–2100.e5.
2. Keirns BH, Koemel NA, Sciarrillo CM, et al. Exercise and intestinal permeability: another form of exercise-induced hormesis? Am J Physiol Gastrointest Liver Physiol 2020;319:G512–G518.
3. Khalili H, Ananthakrishnan AN, Konijeti GG, et al. Physical activity and risk of inflammatory bowel disease: prospective study from the Nurses' Health Study cohorts. BMJ 2013;347:f6633.
4. Zhang Z, Chen L, Qian ZM, et al. Residential green and blue space associated with lower risk of adult-onset inflammatory bowel disease: findings from a large prospective cohort study. Environ Int 2022;160:107084.
5. Miskiewicz A, Kowalczyk P, Oraibi SM, et al. Bird feathers as potential sources of pathogenic microorganisms: a new look at old diseases. Antonie Van Leeuwenhoek 2018;111:1493–1507.
6. Kugathasan S, Saubermann LJ, Smith L, et al. Mucosal T-cell immunoregulation varies in early and late inflammatory bowel disease. Gut 2007;56:1696–1705.
7. Strachan DP. Hay fever, hygiene, and household size. BMJ 1989;299:1259–1260.
8. Kelsen JR, Wu GD. The gut microbiota, environment and diseases of modern society. Gut Microbes 2012;3:374–382.
9. Song SJ, Lauber C, Costello EK, et al. Cohabiting family members share microbiota with one another and with their dogs. eLife 2013;2:e00458.
10. Jiang C, Cui Z, Fan P, Du G. Effects of dog ownership on the gut microbiota of elderly owners. PLoS One 2022;17:e0278105.
11. Santiago Badenas A, Libertucci J, Turpin W, et al. A210: Adlercreutzia is depleted in UC patients, even before clinical diagnosis. J Can Assoc Gastroenterol 2020;3(Suppl 1):82–83.
12. Leibovitzh H, Lee SH, Xue M, et al; CCC GEM Project Research Consortium. Altered gut microbiome composition and function are associated with gut barrier dysfunction in healthy relatives of patients with Crohn's disease. Gastroenterology 2022;163:1364–1376.e10.
13. Barberán A, Dunn RR, Reich BJ, et al. The ecology of microscopic life in household dust. Proc Biol Sci 2015;282:20151139.
14. Gupta S. Microbiome: puppy power. Nature 2017;543:S48–S49.
15. Hua X, Lopes EW, Burke KE, et al. Smoking behavior changes after diagnosis of inflammatory bowel disease and risk of all-cause mortality. J Crohns Colitis 2022;16:1030–1038.
16. Nos P, Domènech E. Management of Crohn's disease in smokers: is an alternative approach necessary? World J Gastroenterol 2011;17:3567–3574.



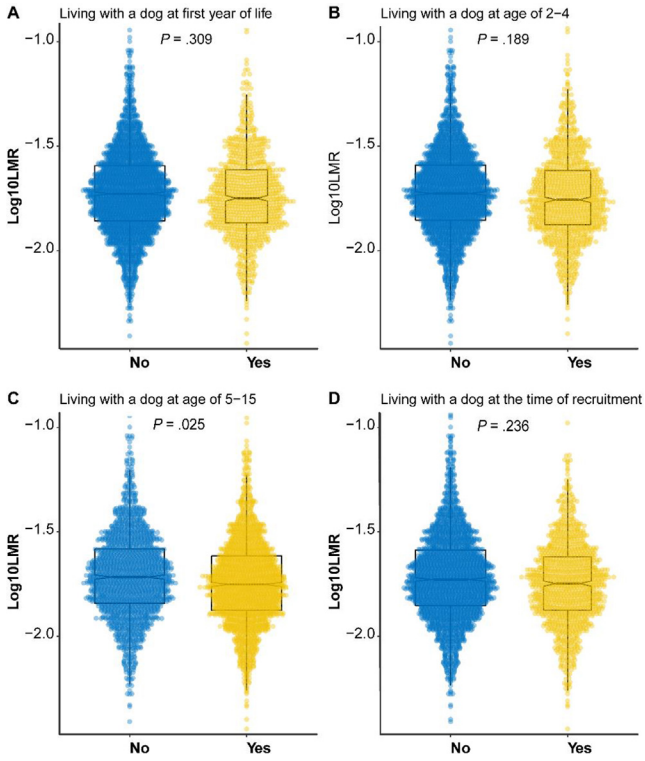
Supplementary Figure 1. Flowchart of exclusion of participants in the analysis of environmental factors related to the onset of CD. [1]A valid CD diagnosis is one that was declared by the subject at their six months follow-up call and verified by their diagnosing physician. [2]Pre-CD individuals are healthy subjects at recruitment, who later developed CD.



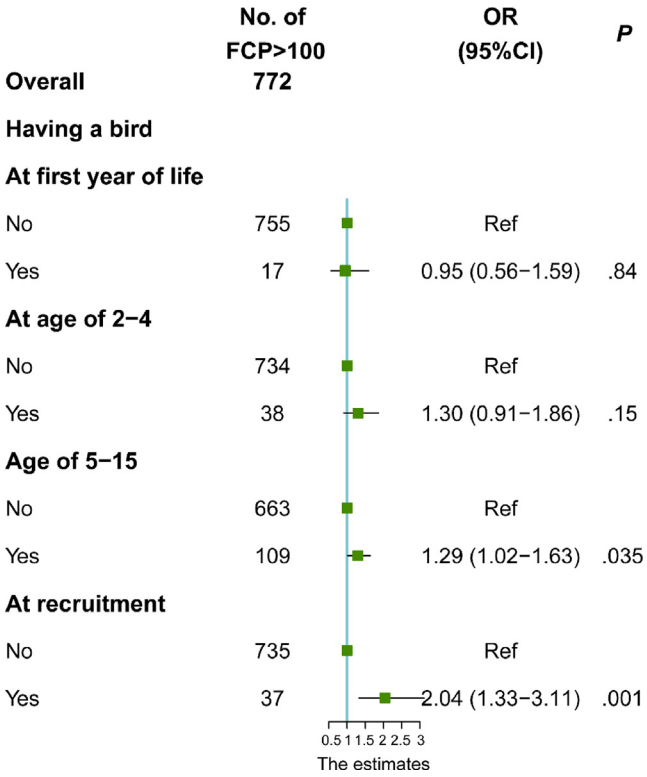
Supplementary Figure 2. Flowchart of environmental variables excluded from the analysis. [1]Descriptive variable included: participant's place of birth, postal code or ZIP code, the name of all medication and supplements that are currently taken on a regular basis, etc.



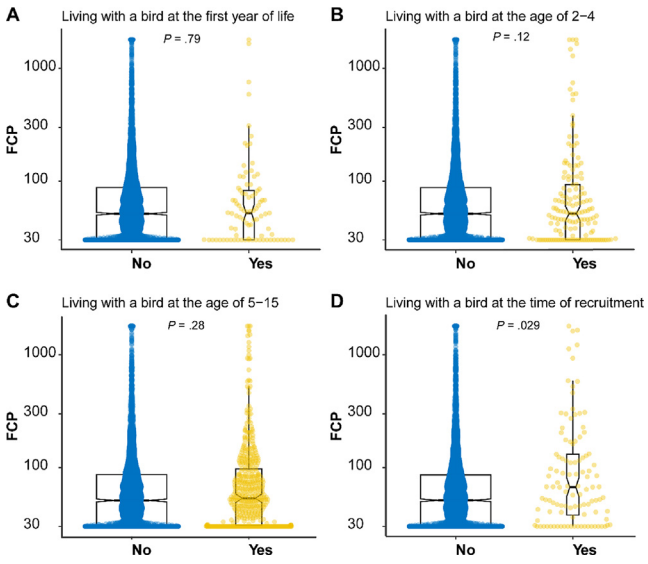
Supplementary Figure 3. Association between exposures to dog and gut permeability (LMR >0.025). Forest plot of the association of participants living with a dog at different age groups with dichotomous LMR. Squares indicate the value of OR, whereas horizontal lines represent the 95% CI of OR; Generalized estimating equation logistic regression accounts for family clustering and further adjusted for age and sex was used.



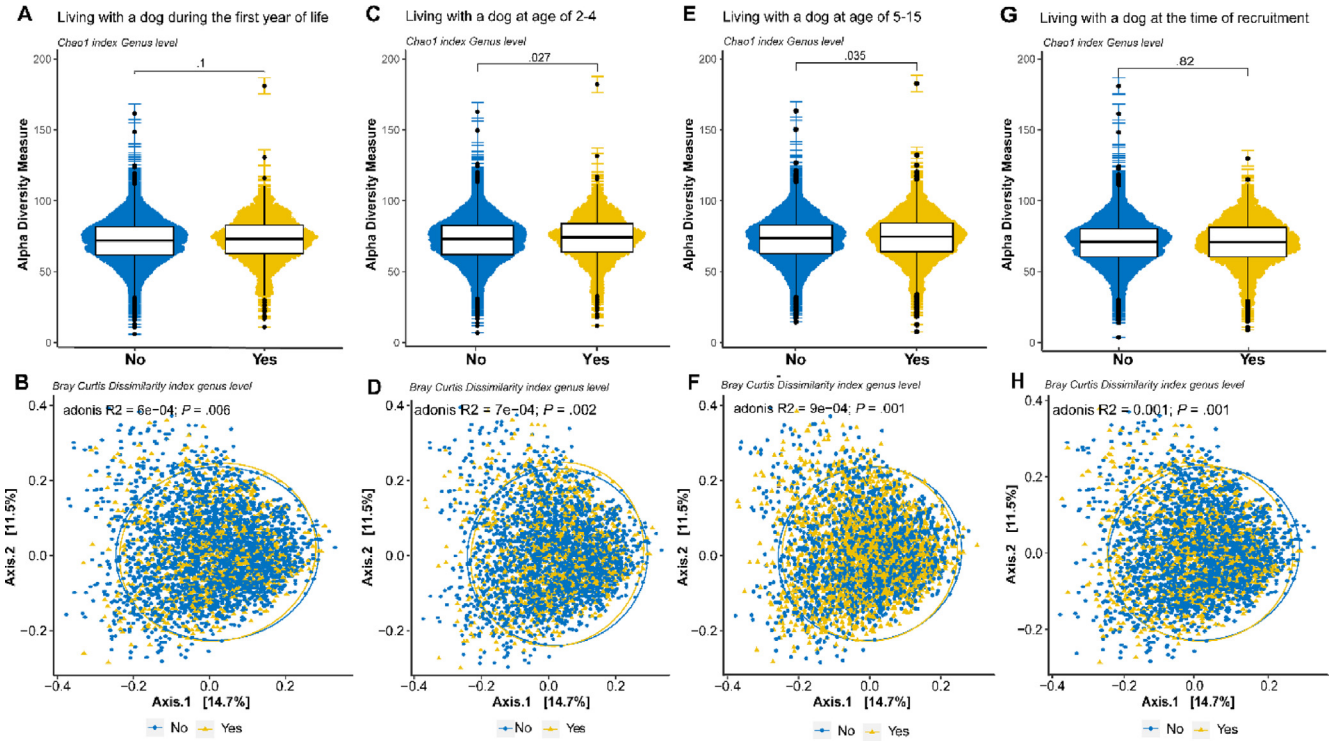
Supplementary Figure 4. Distribution of LMR in participants living with a dog in different age groups. X-axis represents the participants living with a dog: No (*blue*), Yes (*yellow*). Y-axis represents the value of $\log_{10}$ LMR. The lines in the boxplot represent the first, second, and third quartiles. Each dot represents the $\log_{10}$(LMR) for a given participant. The *P*-value generated by generalized estimating equation linear regression accounts for family clustering and further adjusted for age and sex.



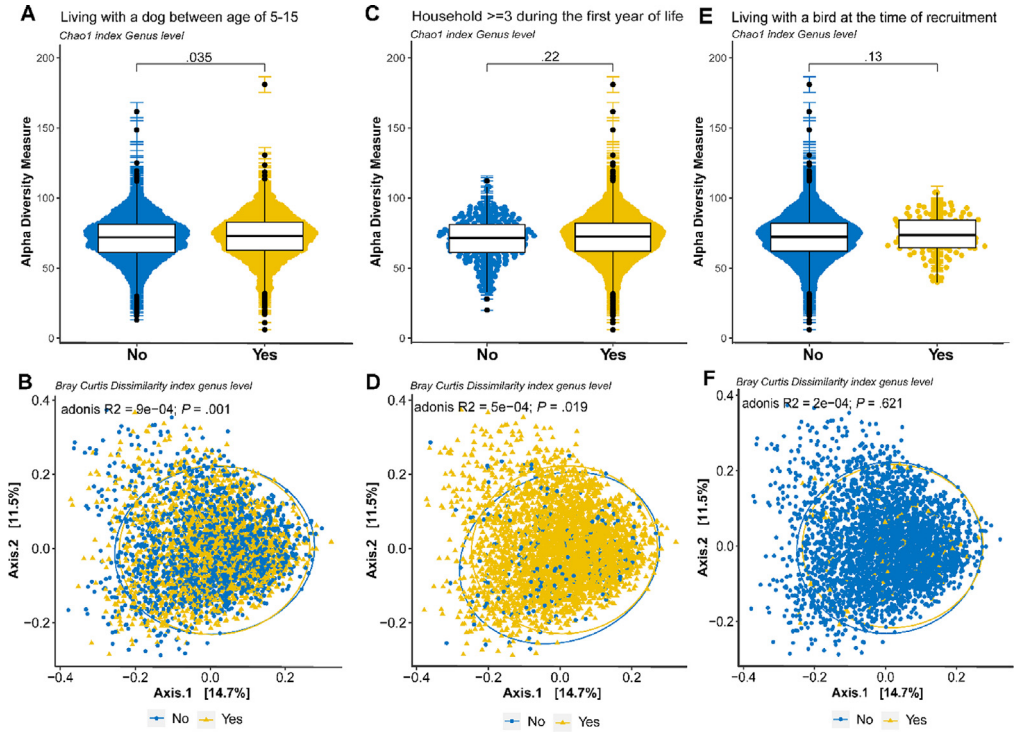
Supplementary Figure 5. Association between participants living with a bird and gut inflammation (FCP >100 μg/g). The forest plot represents the associations of participants living with a bird at different age groups with FCP. Squares indicate the value of OR, whereas horizontal lines represent the 95% CI of OR; Generalized estimating equation logistic regression accounts for family clustering, and further adjustment for age and sex was used.



Supplementary Figure 6. Distribution of FCP in participants living with a bird in different age groups. X-axis indicates participants living with a bird: No (*blue*), Yes (*yellow*). Y-axis represents the value of FCP, Y-axis is $\log_{10}$ transformed for visualization. The lines in the boxplot represent the first, second, and third quartiles. Each dot represents the FCP for a given participant. The *P*-value generated by generalized estimating equation linear regression accounts for family clustering and is further adjusted for age and sex.



Supplementary Figure 7. Association between microbial alpha and beta diversity and participants living or not living with a dog at different age groups. (*A*) Microbial alpha diversity of participants living with a dog vs not living with a dog during the first year of life. (*B*) Microbial beta diversity of participants living with a dog vs not living with a dog during the first year of life. (*C*) Microbial alpha diversity of participants living with a dog vs not living with a dog at age 2 to 4 years. (*D*) Microbial beta diversity of participants living with a dog vs not living with a dog at age 2 to 4 years. € Microbial alpha diversity of participants living with a dog vs not living with a dog at age 5 to 15 years. (*F*) Microbial beta diversity of participants living with a dog vs not living with a dog at age 5 to 15 years. (*G*) Microbial alpha diversity of participants living with a dog vs not living with a dog at the time of recruitment. (*H*) Microbial beta diversity of participants living with a dog vs not living with a dog at the time of recruitment. Beta diversity was calculated using the Bray-Curtis dissimilarity index at the genus level of taxonomy. Each dot represents an individual participant.



Supplementary Figure 8. Association between microbial alpha and beta diversity and CD-associated environmental exposures. (*A*) Microbial alpha diversity of participants living with a dog vs not living with a dog between the age of 5 and 15 years. (*B*) Microbial beta diversity of participants living with a dog vs not living with a dog between the age of 5 and 15 years. (*C*) Microbial alpha diversity of participants living in a household ≥3 during the first year of life. (*D*) Microbial beta diversity of participants living in a household ≥3 during the first year of life. (*E*) Microbial alpha diversity of participants living with a bird at the time of recruitment. (*F*) Microbial beta diversity of participants living with a bird at the time of recruitment. Alpha diversity was calculated using the Chao1 index at the genus level of the taxonomy. Beta diversity was calculated by the Bray-Curtis dissimilarity index at the genus level of the taxonomy. Each dot represents an individual participant.

Supplemental Table 4. Environmental Exposures are Associated With the Future Onset of CD Independent Having a Sibling With CD

Environmental variables: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
Living with a dog between age 5 and 15	2102 (50.0%)	32 (37.2%)	0.63 (0.41–0.98)	.039
Household ≥3 during the first year of life	3954 (94.1%)	77 (89.5%)	0.40 (0.22–0.80)	.01
Currently living with a bird[a]	128 (3%)	8 (9.3 %)	2.86 (1.38–5.93)	.005

Note: *P*-values were generated from Cox proportional hazard regression model including the above 3 environmental exposures separately, adjusting for family clustering, age, sex and having a sibling with CD (compare with offspring).
Note: Household ≥3 indicate individuals living in household with more than 3 individuals.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.
[a]Currently refers to the time of recruitment when the questionnaire was completed.

Supplementary Table 5. Environmental Exposures are Associated With the Future Onset of CD Independent of Breastfeeding

Environmental variables: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
Living with a dog between age 5 and 15	1953 (50.0%)	28 (34%)	0.53 (0.34–0.84)	.007
Household ≥3 during the first year of life	3680 (94.13%)	74 (90%)	0.45 (0.22–0.95)	.035
Currently living with a bird[a]	123 (3%)	8 (9.8 %)	2.90 (1.39–6.05)	.005

Note: *P*-values were generated from Cox proportional hazards regression model including the above 3 environmental exposures separately, adjusting for family clustering, age, sex and if the participants "were breastfed as a baby exclusively or nearly exclusively for longer than 4 months.".
Note: Household ≥3 indicate individuals living in household with more than 3 individuals.
Note: N = 3985 with all data available.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.
[a]Currently refers to the time of recruitment when the questionnaire was completed.

Supplementary Table 6. Environmental Exposures are Associated With the Future Onset of CD Independent of Antibiotic Exposure

Environmental variables: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
Living with a dog between age 5-15	2102 (50.0%)	32 (37.2%)	0.61 (0.39–0.94)	.027
Household ≥3 during the first year of life	3954 (94.1%)	77 (89.5%)	0.43 (0.21–0.86)	.018
Currently living with a bird[a]	128 (3%)	8 (9.3 %)	2.85 (1.38–5.91)	.005

Note: *P*-values were generated from Cox proportional hazards regression model including the above 3 environmental exposures separately, adjusting for family clustering, age, sex, and if the participants had "antibiotics exposure 3 times last year.".
Note: Household ≥3 indicate individuals living in household with more than 3 individuals.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.
[a]Currently refer to the time of recruitment when the questionnaire was completed.

Supplementary Table 7. Association Between Participants Living With a Dog in Different Age Groups and Risk of CD Onset

Living with a dog: yes	Remain healthy	Developed CD	HR (95%CI)	*P*-value
During the first year of life	1067 (25.4%)	18 (20.9%)	0.76 (0.45–1.29)	.309
Between ages 2-4	1241 (29.5%)	20 (23.3%)	0.71 (0.43–1.18)	.189
Between ages 5-15	2102 (50.0%)	32 (37.2%)	0.61 (0.39–0.94)	.025
At recruitment	1303 (31.0%)	22 (25.6%)	0.75 (0.46–1.21)	.236

Note: *P*-values were generated from Cox proportional hazards regression models that included dog exposure in different age groups individually, adjusting for family clustering, age and sex.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.

Supplementary Table 8. Association Between Participants Living With a Dog in Different Age Groups and Risk of CD Onset After Adjusting for Annual Household Income and Education Level

Living with a dog: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
During the first year of life	897 (25.8%)	18 (24.3%)	0.88 (0.52–1.50)	.647
Between ages 2 and 4	1050 (30.2%)	19 (25.7%)	0.78 (0.46–1.31)	.346
Between ages 5 and 15	1779 (51.2%)	29 (39.2%)	0.62 (0.39–0.99)	.044
At recruitment	1112 (32.0%)	22 (29.7%)	0.87 (0.52–1.43)	.578

Note: *P*-values were generated from Cox proportional hazards regression models that included dog exposure in different age groups individually, adjusting for family clustering, age, sex, annual household income and education level.
Note: N = 3548 with all data available.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.

Supplementary Table 9. Association Between Participants Living With a Bird in Different Age Groups and Future Onset of CD

Living with a bird: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
During the first year of life	99 (2%)	1 (1%)	–	–
Between ages of 2 and 4	170 (4.0%)	6 (7.0%)	1.91 (0.83–4.43)	.13
Between ages of 5 and 15	515 (12.3%)	14 (16.3%)	1.49 (0.84–2.66)	.173
At recruitment	128 (3.0%)	8 (9.3%)	2.84 (1.37–5.9)	.005

Note: *P*-values were generated from Cox proportional hazards regression models that included bird exposure in different age groups individually, adjusting for family clustering, age and sex.
Note: A single participant lived with a bird in the first year and developed CD in later life and thus it was not included in the regression analysis according to uniform exclusion criteria. CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.

Supplementary Table 10. Association Between Participants Living in a Household ≥3 Individuals in Different Age Groups and Future Onset of CD

Household ≥3: yes	Remain healthy	Developed CD	HR (95% CI)	*P*-value
During the first year of life	3954 (94.1%)	77 (89.5%)	0.43 (0.22–0.87)	.019
Between ages of 2 and 4	4095 (97.4%)	84 (97.7%)	0.99 (0.24–4)	.984
Between ages of 5 and 15	4129 (98.2%)	85 (98.8%)	1.23 (0.17–8.89)	.84

Note: *P*-values were generated from Cox proportional hazards regression models that included household ≥3 exposure in different age groups individually, adjusting for family clustering, age and sex.
Note: Household ≥3 indicate individuals living in household with more than 3 individuals.
CD, Crohn's disease; CI, confidence interval; HR, hazard ratio.

Supplementary Table 11. Association Between CD-associated Environmental Exposures and Continuous LMR

Identified CD-associated environmental factors: yes	Estimate ± SD	*P*-value
Living with a dog between the age of 5 and 15	−0.02 ± 0.01	.002
Larger family size (≥3) during the first year of life	−0.02 ± 0.02	.290
Currently living with a bird[a]	−0.03 ± 0.02	.054

Note: Univariate analysis of living with CD associated environmental exposures and log(LMR). Generalized estimating equation linear regression accounts for family clustering, and further adjustment for age and sex was used.
CD, Crohn's disease; LMR, lactulose to mannitol ratio; SD, standard deviation.
[a]Currently refers to the time of recruitment when the questionnaire was completed.

Supplementary Table 13. Association Between CD-associated Environmental Exposures and FCP

Identified CD associated environmental factors: yes	Estimate ± SD	*P*-value
Living with a dog between age of 5 and 15	−2.98 ± 7.1	.680
Larger family size (≥3) during the first year of life	8.69 ± 12.13	.470
Currently living with a bird[a]	68.88 ± 31.56	.029

Note: Univariate analysis of living with CD associated environmental exposures and FCP.
Note: Generalized estimating equation linear regression accounts for family clustering, and further adjustment for age and sex was used.
CD, Crohn's disease; FCP, fecal calprotectin; SD, standard deviation.
[a]Currently refers to the time of recruitment when the questionnaire was completed.

Supplementary Table 12. Association Between Participants Living With a Dog in Different Age Groups and Continuous LMR

Living with a dog: yes	Estimate ± SD	*P*-value
During the first year of life	−0.02 ± 0.01	.018
Between ages of 2 and 4	−0.02 ± 0.01	.038
Between ages of 5 and 15	−0.02 ± 0.01	.002
At recruitment	−0.04 ± 0.01	1^{e-05}

Note: Univariate analysis of living with a dog and log (LMR) across different age groups. Note: Generalized estimating equation linear regression accounts for family clustering and further adjusted for age and sex was used.
CD, Crohn's disease; LMR, lactulose to mannitol ratio; SD, standard deviation.

Supplementary Table 14. Association Between Participants Living With a Bird at Different Age Groups and FCP

Having a bird: yes	Estimate ± SD	*P*-value
During the first year of life	7.79 ± 29.09	.790
Between ages of 2 and 4	40.96 ± 26.26	.120
Between ages of 5 and 15	12.87 ± 11.86	.280
At the time of survey	68.88 ± 31.56	.029

Note: Univariate analysis of living with bird and FCP across different age groups.
Note: Generalized estimating equation linear regression accounts for family clustering, and further adjustment for age and sex was used.
CD, Crohn's disease; FCP, fecal calprotectin; SD, standard deviation.